



Learning Grasp Success Prediction from 3D Point Clouds

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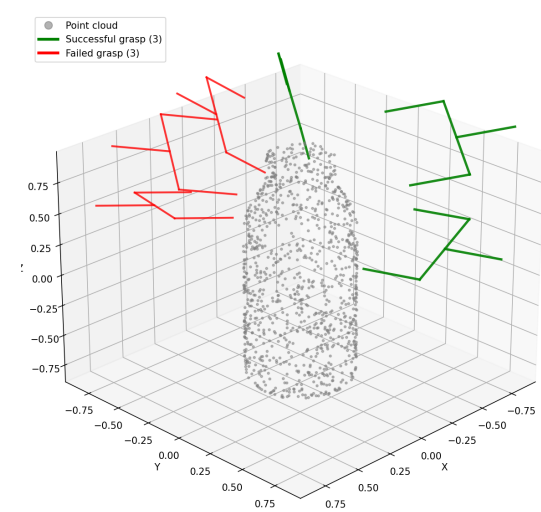
Motivation

Robots need to predict grasp success *before* attempting a grasp. Failed grasps waste time, can damage objects, and may be dangerous. Given a 3D point cloud of an object and a proposed gripper pose, can we predict whether the grasp will succeed?

Problem Setup

Input features: 3D point cloud ($N \times 3$) + grasp pose (position 3D, rotation 9D, width 1D)

Output: Binary success/failure prediction



Same object, different grasps: green=success, red=failure.

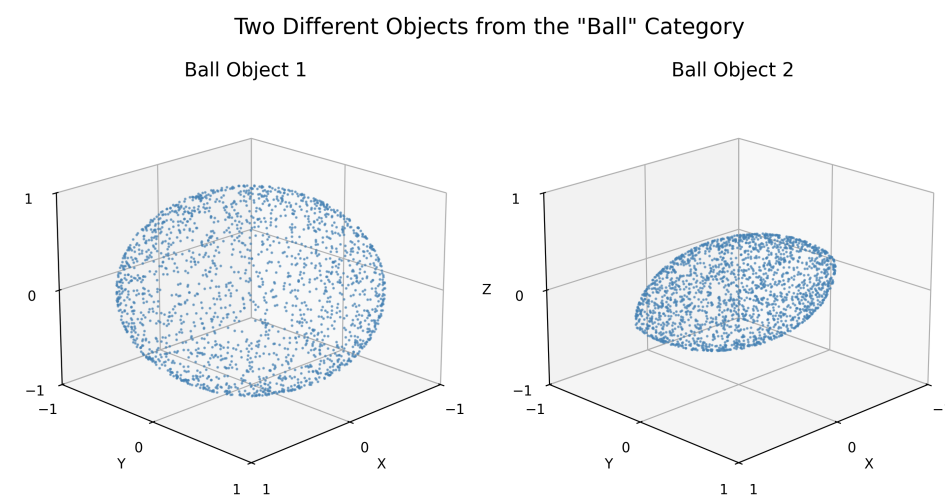
Dataset: ACRONYM

Scale: 17.7M grasp annotations on 8,872 ShapeNet meshes

Class balance: 65% failure, 35% success (imbalanced)

Split: By object category (70/15/15) to test generalization

Our subset: 618K training, 132K test samples (100 grasps/object)



Diversity within one object category ("Ball").

Preprocessing

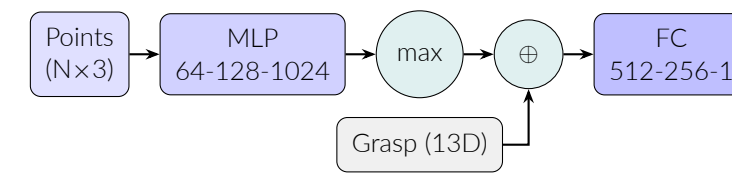
Sample 512–2048 points per mesh.

Normalize points to $[-1, 1]^3$; apply same transform to grasp positions.

Baselines

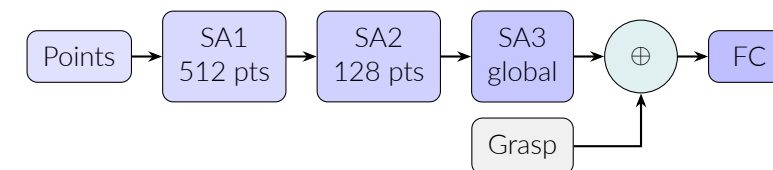
PointNet

Global max-pool treats all points equally.



PointNet++

Hierarchical local features via set abstraction layers.

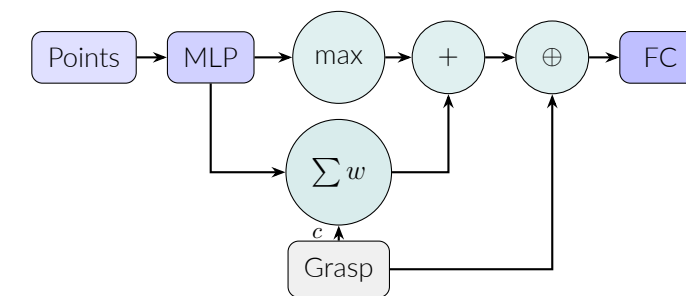


Our Approach: Grasp-Aware Attention

Intuition: Points near the gripper could matter most for predicting contact success.

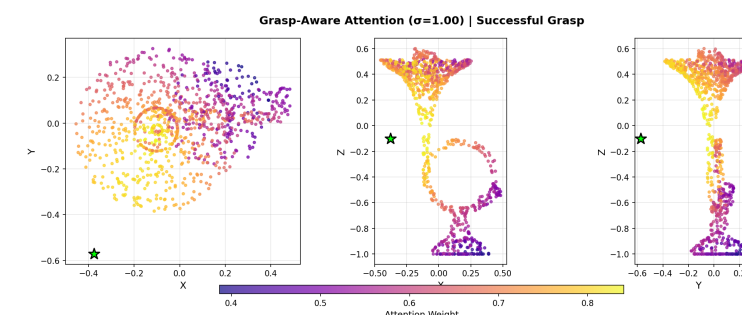
Gaussian Attention

Weight by distance to grasp center c : $w_j = \exp\left(-\frac{\|p_j - c\|_2^2}{2\sigma^2}\right)$

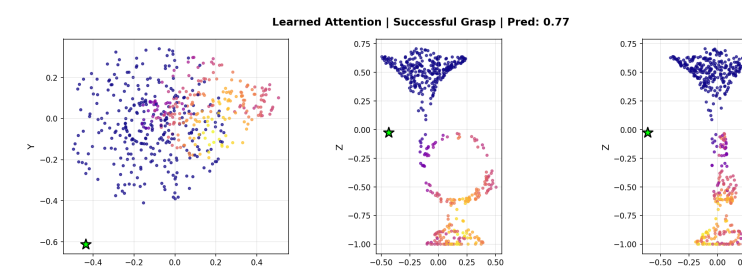


Learned Attention

Query-key attention: $\text{score}_j = q^T k_j / \sqrt{d}$



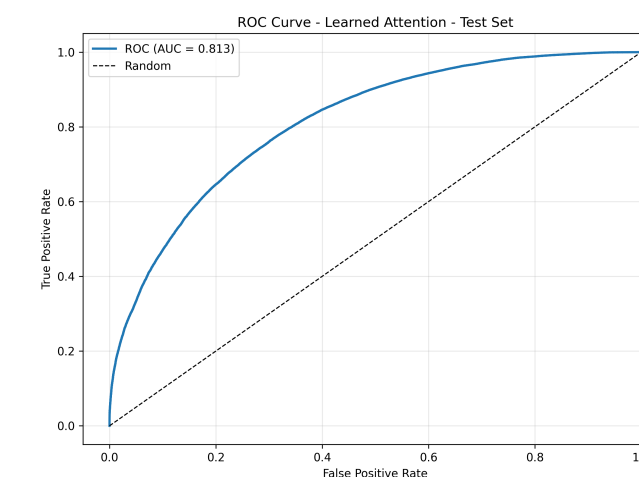
Gaussian: focuses near grasp as intended (yellow=high).



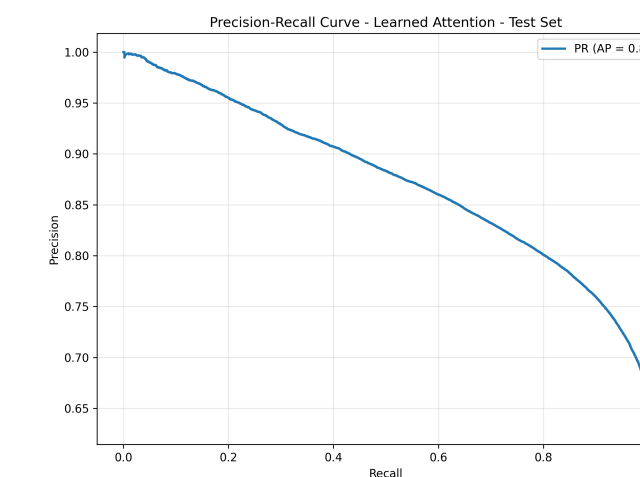
Learned: does not necessarily focus near grasp. Shortcut learning?

Results

Model	Acc	AUC	AP
PointNet	72.9	.795	.864
PointNet++	72.4	.768	.841
Gauss ($\sigma=1$)	73.2	.803	.871
Gauss ($\sigma=2$)	72.1	.796	.866
Learned	73.4	.804	.871
+Reg	73.7	.806	.873
+2048pts	74.1	.813	.875



ROC curve (AUC=0.813)



PR curve (AP=0.875)

Training Setup

Loss: Weighted BCE ($w_+ = 0.52$) · **Optimizer:** Adam, $\text{lr}=10^{-3}$, halve every 20 epochs · **Regularization:** Dropout 0.3–0.5, weight decay 10^{-4} · **Epochs:** 50 with early stopping · **Hardware:** AWS GPU

Ablation Insights

What we learned from experiments:

Attention radius matters: $\sigma=1.0$ beats $\sigma=2.0, 3.0$. Tighter focus on grasp region is better.

More points = more accuracy: 2048 pts gives +0.7% over 1024 pts. Denser geometry helps.

Regularization helps: Dropout 0.5 + weight decay gives +0.3% accuracy and reduces train/val gap.

Surprising findings:

PointNet++ underperforms PointNet. Hierarchical local features don't help; grasp success may depend more on global shape than fine-grained local geometry.

Learned attention doesn't focus near grasp despite achieving best accuracy. Possible shortcut learning exploiting dataset correlations (e.g., grasps near object centers succeed more).

Limitations

Simulation vs. reality: Labels from physics simulation ignore friction, weight, gripper dynamics.

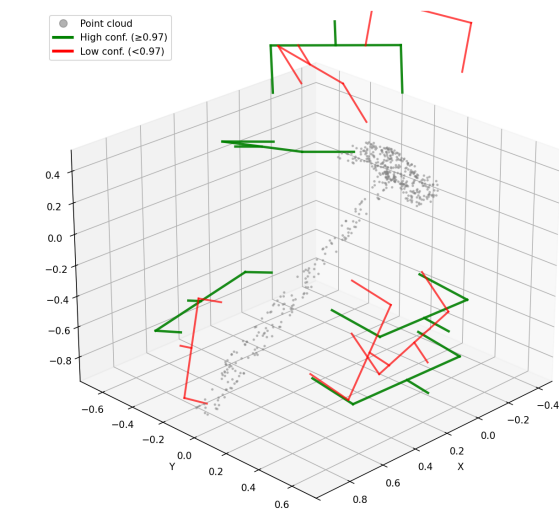
74% may not suffice for real-world deployment where failed grasps are costly.

Parallel-jaw only: Does not generalize to multi-finger or suction grippers.

Extension: Grasp Generation

Heuristic Sampling

Sample grasp candidates along surface normals, reject collisions, rank with our classifier.



Heuristic grasps ranked by classifier. Green=predicted success.

Conditional Diffusion (DDPM)

Trained a denoiser conditioned on point cloud via cross-attention. Grasp as 10D vector (position, 6D rotation, width). Converges but grasps often miss surfaces, requiring geometric constraints.

Conclusion

Main result: Grasp-aware attention achieves **74.1% accuracy, 0.813 ROC-AUC**, outperforming PointNet/PointNet++ baselines.

Takeaway: Gaussian attention is interpretable and focuses near the grasp. Learned attention performs slightly better but may exploit dataset biases (e.g., grasps near object centers succeed more).

Future: Combine attention with hierarchical features; add collision/contact constraints to diffusion.

References

- [1] C. Eppner et al., "ACRONYM: A Large-Scale Grasp Dataset," ICRA 2021.
- C. Qi et al., "PointNet: Deep Learning on Point Sets," CVPR 2017.
- C. Qi et al., "PointNet++: Deep Hierarchical Feature Learning," NeurIPS 2017.